COFFEE FRANCHISE SUPPLY CHAIN MANAGEMENT SYSTEM

Optimizing Franchise Operations

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PAIN POINTS IN CURRENT SYSTEM

I. Supply Chain & Inventory Challenges

- Lack of real-time inventory tracking
- Manual stock management processes
- No automatic low-stock alerts
- Difficulty managing ingredients across franchises

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- Inventory mismatch & stock shortages
- Wasted resources or overstocking
- Delays in fulfilling orders

PAIN POINTS IN CURRENT SYSTEM

II. Order & Operation Challenges

- Inefficient order processing (online & POS not synchronized)
- Limited visibility of live orders
- Manual handling of order status updates
- Difficulty estimating preparation time

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- Order delays & long waiting times
- Increased human errors
- Poor customer experience

PAIN POINTS IN CURRENT SYSTEM

III. Campaign, Promotion & Concurrency Challenges

- High concurrency during flash sale campaigns
- Race conditions in voucher redemption
- Lack of atomic operations in traditional systems

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- Duplicate or invalid voucher redemption
- Data inconsistency
- System instability under heavy load

PAIN POINTS IN CURRENT SYSTEM

IV. Staff & Shift Management Challenges

- Manual shift scheduling
- Lack of coordination between staff
- Difficulty assigning tasks efficiently

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- Operational inefficiencies
- Staff overload or underutilization
- Reduced productivity

PROPOSED SOLUTION

Build a centralized and scalable platform to unify franchise operations

Enable real-time visibility across inventory, orders, and delivery

Automate order processing and stock management workflows

Ensure data consistency and system reliability under high load

Provide integrated analytics for better business decisions

KEY FEATURES / CORE FUNCTIONALITIES

1.User Management

- Manage user accounts
- Assign roles and permissions

2.Product Management

- Manage categories and products catalog (create, update, delete)
- Define pricing and availability

3. Sales & Order Management

- Online Ordering via web.
- POS (In-store Sales) for staff to take customer orders
- Customers can track order and delivery status.

4. Supply Chain

- Inventory management for ingredients and materials.
- Bill of Materials (BOM) to map products to required ingredients.
- Automatic stock deduction.

5.Shift Management

- Shift planning for employees across stores.
- Staff management and role assignment.
- Internal delivery management and order status updates.

6.Dashboard

- Real-time overview of sales and orders.
- KPI tracking (revenue, orders, performance).
- Detailed statistics on all orders.

7.Voucher Management

- Create and manage discount vouchers.
- Configure usage rules and validity.
- Launch Flash Sale ensure during high traffic

8.Franchise Management

- Manage multiple branches/franchises.
- Monitor branch-level operations.
- Transfer branch/franchise

9.Notification

- OTP register account.
- Email order status.
- Email transfer contract.

10.Customer Engagement

- Loyalty programs with points and customer tiers.
- Promotion engine supporting complex discount rules.
- Coupon and voucher management.

BUSINESS VALUE

BUSINESS VALUE

1. Value Proposition

Delivering a seamless omnichannel experience with offline-capable AI ordering and end-to-end transparency.

2. Franchisor

Empowering brand growth through centralized marketing, dynamic promotions, and data-driven loyalty

3. Franchise

Optimizing store profitability with smart inventory tracking and automated cost controls

4.Tech/Architecture

Built on a resilient, scalable architecture to streamline store operations and network expansion

Value Proposition

Current Problem

- Franchise chains often face difficulties in data synchronization, remote quality control, and supply chain management when expanding to multiple locations.

CFMS Solution

- A centralized management platform connecting the franchisor and all franchise stores in one unified system.

Operational Excellence

- Automates ordering, inventory management, and sales processes.

Transparent Control

- Provides real-time insights into revenue, costs, and cash flow.

Sustainable Scalability

- A scalable architecture that supports rapid expansion to hundreds of stores without performance bottlenecks.

Franchisor

Contract Lifecycle Management

- Digitizes the entire franchise contract process including approval, signing, renewal, and royalty fee tracking.

Global Business Visibility

- Provides real-time reports on revenue, cost, and profit across all franchise locations.

Supply Chain Synchronization

- Centralized ingredient pricing managed by the franchisor and automatically synchronized across all stores.

Financial Transparency

- Centralized wallet transaction tracking and automated reconciliation of royalty fees.

Franchise

Smart Inventory Management

- Tracks stock levels accurately and calculates inventory value based on historical purchase prices.

Automated Cost Calculation

- Uses recipe formulas and real-time ingredient prices to automatically calculate the Cost of Goods Sold (COGS).

Seamless Sales Operations

- Smooth order creation, flexible payment processing, and efficient refund handling.

Operational Insights Dashboard

- Provides store owners with dashboards showing sales trends, payment history, and operational performance.

Tech/Architecture

High Availability & Service Isolation

- If one service is under maintenance (e.g., Supplier Service), other services such as Sales can continue operating normally.

Cost-Efficient Scaling

- Services can scale independently.
- For example, SalesService can scale during peak hours without scaling the entire system.

Data Consistency with Event-Driven Architecture

- Complex operations such as refunds, inventory updates, and wallet transactions are handled safely using event-driven workflows.

High Code Quality

- Over 80% test coverage using xUnit, Moq, and FluentAssertions, validated through SonarQube to ensure system stability and maintainability.

COMPLEXITY & CHALLENGES

CFMS is not just a standalone sales software. It is an Enterprise ecosystem specifically designed to handle massive data storage volumes and the complex operational challenges of the Franchise chain model.

1. Infrastructure & Data Scale Challenges

Ensuring data consistency and maintaining millisecond-level response times while synchronizing transactions across dozens of microservices and distributed databases.

2. The Core Domain Puzzle

Digitizing and fully automating the most complex F&B operations, including intelligent shift scheduling, cross-location attendance tracking, overnight shift management, and sophisticated payroll rules.

3. UX & Cost Optimization Pressure

Handling massive volumes of visual data directly on the web (avoiding lag), while integrating proprietary routing technology to shield investors from the high costs of third-party APIs.

COMPLEXITY & CHALLENGES

To address these challenges, CFMS leverages five key cutting-edge technologies:

- **Distributed Microservices Architecture**
- **Real-time Event-Driven Architecture**
- **Greedy Algorithm for Shift Scheduling**
- **Next-Generation Massive Data Interaction Interface**
- **Logistics Infrastructure & Operational Optimization (OSRM)**

DISTRIBUTED MICROSERVICES ARCHITECTURE

Complexity:

- A large-scale system fragmented into 10+ independent microservices (Sales, Franchise, Inventory, Identity, etc.)
- Data is distributed across multiple databases (Polyglot Persistence), rather than stored in a single location

Challenges:

- Maintaining data consistency when a workflow spans across 3–4 services
- Efficiently routing millions of secure requests through the API Gateway without creating bottlenecks

System Solution:

- Apply Domain-Driven Design (DDD) to clearly separate business domains
- Use a centralized API Gateway for orchestration
- Fully containerize services using Docker

REAL-TIME & ASYNCHRONOUS DATA PROCESSING

Complexity:

- Highly interwoven business flows: a successful order instantly triggers a chain of events [Inventory Deduction] + [Revenue Update] + [Notification Dispatch]
- Processing speed must achieve millisecond-level latency (near zero-latency)

Challenges:

- Handling bottlenecks during peak times such as flash sales or holidays in the F&B sector
- Ensuring no message loss and preventing duplicate processing (idempotency), avoiding issues like double charges or incorrect inventory updates

Infrastructure Solution:

- Move beyond traditional queues by leveraging Apache Kafka and Zookeeper for event streaming and message orchestration

AUTOMATING THE “OPERATIONAL HR BRAIN” (HR CORE)

Complexity:

- Real-world F&B labor rules are highly strict: overnight shifts spanning two days, maximum working hours per week, and distinctions between part-time and full-time employees
- Complex shift swap mechanisms between employees, along with flexible check-in/out grace periods

Challenges:

- Designing logic to eliminate “schedule conflicts” across all employees
- Building an algorithm that scans the entire workforce and fills shifts naturally without overloading CPU/server resources

Algorithmic Solution:

- Pioneering the use of a Greedy Algorithm to automatically evaluate skills and assign the most optimal staffing decisions in real time

NEXT-GENERATION GRAPHICAL INTERFACE EXPERIENCE (UI/UX)

Complexity:

- Rendering massive Big Data directly in the web browser (dense FullCalendar schedules, multi-stream dashboards)
- Embedding advanced visual interactions and 3D graphics for spatial management

Challenges:

- Managing complex client-side state (static/dynamic) to ensure synchronization with the server
- Optimizing performance to maintain smooth rendering (60 fps), even on low-end cashier devices

Algorithmic Solution:

- Built with Next.js 16
- State management powered by Zustand & React Query
- Integrated 3D graphics using Three.js

LOGISTICS & OPERATIONAL COST OPTIMIZATION

Complexity:

- F&B chains often suffer cost leakage due to heavy reliance on Google Maps services and loosely integrated payment gateways

Challenges:

- Building an in-house Routing Engine requires deep infrastructure expertise and significant resources
- Ensuring top-level encryption and security for real-time Webhook handling from VNPay via secure protocols

Algorithmic Solution:

- Develop a proprietary distance calculation system using OSRM containers
- Integrate Ngrok tunneling for payments and SonarQube for continuous core system security scanning

FUTURE IMPROVREMENT

FUTURE FEATURES

Live Order Tracking (Delivery Phase)

- Real-time Live Order Tracking during delivery
- Display delivery status updates (preparing → delivering → completed)
- Provide ETA updates for customers and staff
- Enable driver tracking (optional GPS integration)
- Notify users of delays or issues in delivery

FUTURE FEATURES

POS Order Management Service

- Implement POS Order Entry for direct in-store order creation
- Allow staff to Update Order Status in real time
- Provide Live Order View for monitoring active orders
- Add Preparation Time Estimation for better planning
- Enable Order List View for tracking all orders
- Support Edit Order functionality after creation

THANK YOU FOR LISTENING!

Questions & Answers